

**This Is Not the ex-Gaussian Model You Are Looking For: On the Default Parameterization of  
Bayesian ex-Gaussian Models in brms**

Jason Geller<sup>1</sup> and Bernhard Angele<sup>2</sup>

<sup>1</sup>Department of Psychology and Neuroscience, Boston College

<sup>2</sup>Nebrija University

**Author Note**

Jason Geller  <https://orcid.org/0000-0002-7459-4505>

Bernhard Angele  <https://orcid.org/0000-0001-8989-8555>

The analyses herein were not preregistered. Data, code, and materials for this manuscript can be found at the Github The authors have no conflicts of interest to disclose.

Author roles were classified using the Contributor Role Taxonomy (CRediT; <https://credit.niso.org/>) as follows: *Jason Geller*: conceptualization, data curation, formal analysis, project administration, resources, visualization, and writing – original draft. *Bernhard Angele*: formal analysis, validation, writing – original draft, and Writing - review & editing

Correspondence concerning this article should be addressed to Jason Geller, Department of Psychology and Neuroscience, Boston College, McGuinn 300, Chestnut Hill, MA 2467, USA, Email: [drjasongeller@gmail.com](mailto:drjasongeller@gmail.com)

**Abstract**

The ex-Gaussian distribution is a popular mathematical model for analyzing response time (RT) data because it separates the central portion of the distribution, captured by the Gaussian parameters  $\mu$  and  $\sigma$ , from the positively skewed tail, captured by the exponential parameter  $\tau$ . Researchers often use these parameters to make inferences about underlying cognitive processes. As Bayesian methods have become increasingly accessible, complex ex-Gaussian models can now be fit with relative ease, and in R, the {brms} package makes this especially straightforward. However, it is important to recognize that the default parameterization offered by {brms} does not map onto the classical parameterization familiar to many researchers in experimental psychology in the default model,  $\mu$  indexes the mean of the full ex-Gaussian distribution rather than the location of the Gaussian component alone. This distinction matters because changes in the Gaussian location and changes in the exponential tail can offset one another at the level of the overall mean, so some published studies may have drawn incorrect inferences from effects estimated on the default  $\mu$  parameter. Using simulated data and a published finding we highlight how using the default parameterization can lead to incorrect inferences. We then demonstrate how to fit the classical parameterization directly in {brms}, with the aim of helping researchers and reviewers align ex-Gaussian model parameters with their theoretical questions.

*Keywords:* ex-Gaussian, reaction time, Bayesian estimation, brms, parameterization

## **This Is Not the ex-Gaussian Model You Are Looking For: On the Default Parameterization of Bayesian ex-Gaussian Models in brms**

Over the past two decades, psychology has seen a substantial increase in the application of Bayesian data analysis methods as a supplement to widely used frequentist methods (Pfadt et al., 2025). This shift has been driven by the increasing availability of computational tools that have made fitting Bayesian models both practical and efficient. Programming languages like R (R Core Team, 2024), Python (Van Rossum & Drake, 2009), and Julia (Bezanson et al., 2017) provide libraries and packages that allow researchers to fit Bayesian models with little technical overhead. In addition, growing dissatisfaction with null-hypothesis significance testing (NHST) and its associated pitfalls (e.g., p-value misinterpretation and dichotomous thinking), together with the replication crisis/credibility revolution (Vazire, 2018), has pushed many researchers toward statistical alternatives.

One domain that has seen increased adoption of Bayesian methods is experimental psychology, where one of the most common dependent variables is response time (RT) data, which is a measure of how long it takes a person to initiate some type of response (e.g., motor, written, or speech). RT data offers a window into cognitive processes underlying behavior (Baayen & Milin, 2010). Traditionally, RT data have been analyzed using summary statistics such as the mean or median. However, RT distributions are typically positively skewed, bounded at zero, and characterized by substantial variability both within and between individuals (Whelan, 2008; Wilcox & Keselman, 2003). Moreover, experimental manipulations may affect not only the central tendency of the distribution but also its variability or upper tail. Consequently, analyses based solely on the mean or median can obscure theoretically meaningful distributional effects, motivating calls to move beyond mean-based analyses of RT data (Balota & Yap, 2011).

One way to move beyond the mean is to model the full shape of the RT distribution. Mathematical models provide a flexible way to do so, allowing researchers to draw inferences about the psychological processes—such as memory, attention, and language—that give rise to observed response times. One model that has become especially popular for this purpose is the ex-Gaussian model (Ratcliff & Murdock, 1976). The ex-Gaussian distribution is formed by the convolution of a Gaussian distribution and an exponential distribution and is characterized by three parameters:  $\mu$ ,  $\sigma$ , and  $\tau$ . The parameter  $\mu$  represents the mean of the Gaussian component, whereas  $\sigma$  represents its standard deviation. The  $\tau$  parameter represents both the mean and standard deviation of the exponential component and primarily governs the length of the distribution's positively skewed upper tail. The overall mean of the ex-Gaussian

distribution is  $\mu + \tau$ , and the standard deviation (SD) is  $\sqrt{\sigma^2 + \tau^2}$ . Although these parameters have been used to infer cognitive processes such as automaticity (Balota et al., 2008; Dewitt et al., 2018) and cognitive control/inhibition (Kane & Engle, 2003), these interpretations are not inherent to the parameters and should be made cautiously (Matzke & Wagenmakers, 2009). Nevertheless, the ex-Gaussian model can provide a more complete description of RT data than the arithmetic mean alone, which can be disproportionately influenced by the positively skewed tail of the distribution.

The ex-Gaussian model can be fit within a Bayesian framework relatively easily using the {brms} package in R which provides a high-level interface to the probabilistic programming language Stan (Carpenter et al., 2017). The {brms} package supports a wide range of response distributions, including the ex-Gaussian family, and allows researchers to incorporate complex model structures at both the participant and item levels as well as across different parameters. This accessibility has contributed to the growing use of Bayesian ex-Gaussian models in reaction-time research (see Angele et al., 2023, 2024; Geller et al., 2025; Kinoshita & Liong, 2023; Serrano-Carot et al., 2026). However, an important complication arises from the default parameterization used by the R package {brms}. Although the model is straightforward to estimate, its default specification does not necessarily provide direct access to the parameter-specific comparisons researchers often wish to make.

The central problem is that the parameter labeled  $\mu$  in the default {brms} ex-Gaussian model does not correspond to the mean of the Gaussian component in the conventional parameterization. Instead, the quantity modeled as  $\mu$  in {brms} represents the mean of the full ex-Gaussian distribution. Under the conventional parameterization, this quantity is equal to  $\mu_{\text{Gaussian}} + \tau$ . Consequently, regression effects estimated for the default  $\mu$  parameter reflect changes in a composite quantity that combines the Gaussian location and exponential tail components. Such effects therefore cannot be interpreted directly as changes in the Gaussian component or in typical response speed. To recover the conventional Gaussian mean, it is necessary to derive it from the model parameters as  $\mu_{\text{Gaussian}} = \mu_{\text{brms}} - \tau$  or to specify an alternative parameterization that models the conventional ex-Gaussian parameters directly.

In this commentary, we demonstrate what the default parameterization of the ex-Gaussian model in {brms} actually provides and then show how to specify the model so that its parameters correspond to the conventional ex-Gaussian parameters used in reaction-time research. The goal is to improve the accuracy and interpretability of inferences drawn from these models. This issue is particularly important because the ex-Gaussian distribution is widely used to distinguish changes in the central portion of the RT distribution from changes in its positively skewed tail.

**Simulation Example**

To demonstrate the issue with fitting the default ex-Gaussian model in {brms}, I simulate data using a comparison adapted from one of my own studies (Geller et al., 2025) in R [version 4.6.0; R Core Team (2024)]. In Geller et al. (2025), we examined how perceptual disfluency—manipulated by varying the level of blur applied to letter strings influenced reaction times in a lexical decision task (i.e., is this letter string a word?), and whether these effects were related to how well someone remembers the words in a recognition memory task. In that study, low blur words affected early stages of encoding, reflected in the Gaussian location parameter  $\mu$ , but not the later, tail component captured by  $\tau$ . For the present illustration, however, I construct a sharper scenario in which  $\mu$  and  $\tau$  differ in opposite directions so that the two conditions share the same expected reaction time. Using the `rexgaussian2()` function from the {brms.exgaussian} package, I simulate ex-Gaussian data for clearly presented and low-blur words in which  $\mu$  is set to 500 ms and 550 ms,  $\tau$  to 100 ms and 50 ms, and  $\sigma$  to 60 ms in both conditions (see Table 1).

```
# install from Github
#remotes::install_github("mattansb/brms.exgaussian")

library(brms.exgaussian)
```

**Table 1**

*Data-generating ex-Gaussian parameters for the Clear and Low-Blur conditions, with opposing changes in  $\mu$  and  $\tau$  that hold the expected response time ( $\mu + \tau$ ) constant.*

| Blur     | $\mu$ | $\sigma$ | $\tau$ | Expected RT |
|----------|-------|----------|--------|-------------|
| Clear    | 500   | 60       | 100    | 600         |
| Low-Blur | 550   | 60       | 50     | 600         |

**Simulate the Data**

We first draw 100 reaction times per condition (200 in total) from the ex-Gaussian distributions defined by these parameters.

```
# Fix the random seed so the simulated data are identical on every render.
set.seed(1234)

# Number of simulated trials per condition (200 observations in total).
n_per_group <- 100

# One row per trial, labelled by Blur condition (Clear is the reference level).
```

```

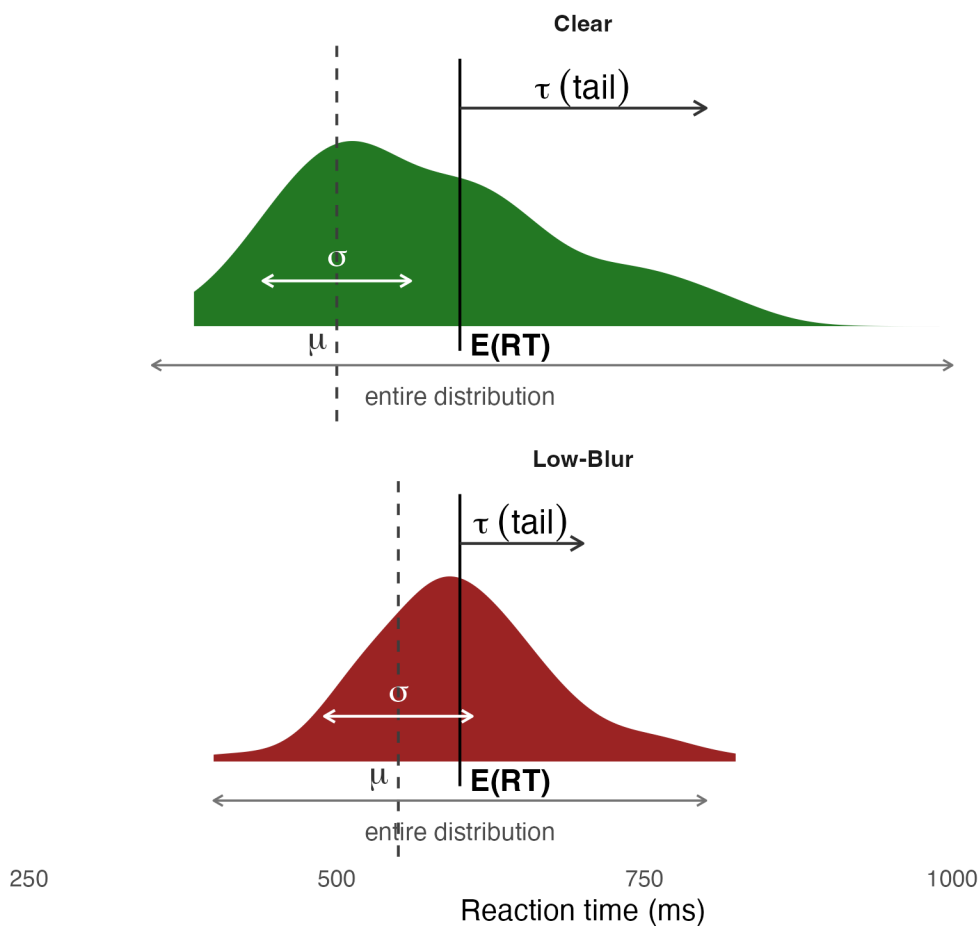
135 d <- data.frame(
136   Blur = factor(
137     rep(c("Clear", "Low-Blur"), each = n_per_group),
138     levels = c("Clear", "Low-Blur")
139   )
140 )
141
142 # Row index (1 = Clear, 2 = Low-Blur) used to look up each trial's parameters.
143 parameter_index <- as.integer(d$Blur)
144
145 # Draw reaction times from the ex-Gaussian using each trial's condition values.
146 d$rt <- brms.exgaussian::rexgaussian2(
147   n = nrow(d),
148   mu = sim_params$mu[parameter_index],
149   sigma = sim_params$sigma[parameter_index],
150   tau = sim_params$tau[parameter_index]
151 )

```

Figure 1 shows the simulated RT distribution for each condition and how the three ex-Gaussian parameters shape it:  $\mu$  locates the Gaussian component,  $\sigma$  sets its spread, and  $\tau$  stretches the right tail. Because the increase in  $\mu$  from Clear to Low-Blur is offset by an equal decrease in  $\tau$ , the overall mean  $E(RT) = \mu + \tau$  is identical (600 ms) in both conditions.

**Figure 1**

Simulated reaction-time distributions for each condition, annotated with the ex-Gaussian parameters. The dashed line marks the Gaussian location  $\mu$ ; the double arrow spans  $\pm\sigma$  around it (the Gaussian spread); the rightward arrow over the tail marks the exponential component  $\tau$ , which stretches the distribution to the right; and the solid line marks the overall mean  $E(RT) = \mu + \tau$ , which is 600 ms in both conditions. The bottom span highlights that  $E(RT)$  is the mean of the entire distribution — the quantity the default brms mu estimates.

**Default {brms} Parameterization**

Using this data we fit the default brms ex-Gaussian model. In this parameterization, the parameter labeled mu represents the expected value of the full ex-Gaussian distribution rather than the location of its Gaussian component.

The exponential component is labeled beta. On the response scale, beta corresponds to the classical ex-Gaussian parameter  $\tau$ :

$$\beta = \tau_{\text{classical}}.$$

Thus,  $\beta$  and  $\tau$  are not substantively different quantities. Both represent the mean of the exponential component. The difference is primarily one of notation. The important difference between the default {} model and the classical parameterization concerns  $\mu$ :

$$\mu = \mu_{\text{classical}} + \tau.$$

In other words, the default {} parameterization uses  $\mu$  for the expected value of the complete exponential-Gaussian distribution, whereas the classical parameterization uses  $\mu$  for the location of the Gaussian component.

Both the expected response and the exponential component are allowed to vary across conditions.

### **Prior Specification**

The priors were selected to reflect the known scale of the simulated data while remaining sufficiently broad to allow the model to recover deviations from the generating values.

```
priors_default <- c(
  # mu intercept: centered at 600 ms, the expected RT (E(RT)) of the reference
  # condition in this default parameterization.
  prior(
    normal(600, 100),
    class = "Intercept"
  ),
  # Effect of Blur on mu: centered at 0 (no assumed direction).
  prior(
    normal(0, 50),
    class = "b"
  ),
  # beta (exponential mean) intercept, on the log scale: exp(log(100)) = 100 ms.
  prior(
    normal(log(100), 0.5),
    class = "Intercept",
    dpar = "beta"
  ),
  # Effect of Blur on beta (log scale): centered at 0 (a ratio of 1).
  prior(
    normal(0, 0.5),
    class = "b",
    dpar = "beta"
  )
)
```

The intercept prior for the primary  $\mu$  submodel is

$$\mu_{\text{Intercept}} \sim \mathcal{N}(600, 100).$$



206 We center this prior at 600 ms because, in the default parameterization,  $\mu$  represents the expected RT.

207 The simulated expected RT for the reference condition is

$$500 + 100 = 600 \text{ ms}.$$

208 The standard deviation of 100 ms makes the prior informative about the plausible scale of RTs without

209 tightly constraining the intercept to equal exactly 600 ms.

210 The prior on the `Blur` coefficient is

$$b_{\text{Blur}} \sim \mathcal{N}(0, 50).$$

211 This prior is centered at zero because, before observing the data, positive and negative condition effects

212 are both plausible. A standard deviation of 50 ms allows substantial differences between conditions while

213 placing less prior mass on implausibly large effects.

214 The beta parameter is positive and uses a log link. Its intercept is therefore modeled on the log

215 scale:

$$\log(\beta_{\text{Clear}}) \sim \mathcal{N}(\log(100), 0.5).$$

216 The prior is centered at  $\log(100)$  because the simulated value of the exponential mean for the reference

217 condition is 100 ms. After applying the inverse-link transformation, the prior is centered approximately

218 around

$$\beta_{\text{Clear}} = 100 \text{ ms}.$$

219 The standard deviation of 0.5 is on the log scale. It therefore represents multiplicative rather than additive

220 uncertainty.

221 The prior on the effect of `Blur` on beta is

$$b_{\beta, \text{Blur}} \sim \mathcal{N}(0, 0.5).$$

222 Because this coefficient is also on the log scale, a value of zero represents no difference between

223 conditions:

$$\exp(0) = 1.$$

224 The simulated change from 100 ms to 50 ms corresponds to the log ratio

$$\log\left(\frac{50}{100}\right) = \log(0.5) \approx -0.693.$$

225 The prior  $\mathcal{N}(0, 0.5)$  places reasonable prior mass near this value while still favoring smaller differences.

226 The model can then be fit as follows:

```

227 fit_brms <- brm(
228   bf(
229     rt ~ Blur,      # mu submodel: here mu is E(RT), not the Gaussian location
230     sigma ~ 1,      # constant Gaussian SD
231     beta ~ Blur     # exponential mean (the classical tau), allowed to vary
232   ),
233   data = d,
234   prior = priors_default,
235   family = exgaussian(), # 's built-in ex-Gaussian family
236   chains = 2,
237   iter = 2000,
238   warmup = 500,
239   backend = "cmdstanr",
240   cores = 4,
241   seed = 1234          # reproducible draws
242 )

```

### 243 *Estimating Ex-Gaussian Parameters with {easystats}*

244 The {modelbased} package, which is part of the {easystats} ecosystem, can easily be used to  
 245 obtain condition-specific posterior estimates from a fitted model. The `estimate_means()` function returns  
 246 estimated parameters for each condition, back-transformed, so estimates are presented on time scale,  
 247 whereas `estimate_contrasts()` computes differences between conditions.

248 As shown in [Table 1](#), the present data were generated so that the two conditions have the same  
 249 expected reaction time but differ in the Gaussian and exponential components of the ex-Gaussian  
 250 distribution.

251 Under the classical ex-Gaussian parameterization, the expected response is

$$E(RT) = \mu + \tau.$$

252 Thus, the 50-ms increase in  $\mu$  for the Low-Blur condition is offset by a 50-ms decrease in  $\tau$ . The expected  
 253 RT is therefore 600 ms in both conditions.

### 254 *Estimating the Expected Response ( $\mu$ )*

255 Condition-specific estimates of the expected response can be obtained with `estimate_means()`  
 256 and setting `predict` argument to “mu”:

```

257 # Default model: per-condition mu, which equals E(RT) in this parameterization.
258 overall_means <- estimate_means(
259   fit_brms,
260   by = "Blur",
261   predict = "mu",
262   centrality = "mean"
263 )

```

**Table 2**

*Estimated expected response,  $E(RT)$ , by condition from the default model.*

| Blur     | Mu    | 95% CrI        | pd |
|----------|-------|----------------|----|
| Clear    | 580.8 | [559.1, 603.8] | 1  |
| Low-Blur | 603.2 | [588, 618.9]   | 1  |

Because the primary mu parameter in the default ex-Gaussian model represents  $E(RT)$ , these estimates should be close to 600 ms in both conditions.

The corresponding contrast can be obtained using `estimate_contrasts()`:

```
# Default model: Blur contrast in E(RT) ( $\approx 0$  by design).
overall_difference <- estimate_contrasts(
  fit_brms,
  contrast = "Blur",
  centrality = "mean"
)
```

**Table 3**

*Blur contrast in the expected response from the default brms model.*

| Contrast         | Mean | 95% CrI      | pd    |
|------------------|------|--------------|-------|
| Low-Blur - Clear | 22.4 | [-4.8, 48.8] | 0.949 |

The posterior estimate of this contrast should be close to zero because the two conditions were generated with the same expected RT:

$$(550 + 50) - (500 + 100) = 600 - 600 = 0.$$

Thus, the default model correctly recovers the absence of a difference in the full distributional mean.

However, its primary mu coefficient does not directly reveal the 50-ms difference in the Gaussian-location parameter.

**Estimating the Exponential Component.**

In the default {brms} parameterization, beta represents the mean of the exponential component. It therefore corresponds directly to  $\tau$  in the classical parameterization:

$$\beta = \tau.$$

Condition-specific estimates can be obtained by requesting predictions for beta:

```
# Default model: per-condition beta (the exponential mean = classical tau).
tau_estimates_default <- estimate_means(
  fit_brms,
```

```

294   by = "Blur",
295   predict = "beta",
296   centrality = "mean"
297 )

```

#### 298 **Table 4**

299 *Estimated exponential component (beta) by condition from the default brms model.*

|     | Blur     | Beta  | 95% CrI       | pd |
|-----|----------|-------|---------------|----|
| 300 | Clear    | 107.3 | [79.1, 139.6] | 1  |
| 301 | Low-Blur | 55.7  | [34.9, 79.5]  | 1  |

303 Although the resulting object is called `tau_estimates_default` for interpretive clarity, the  
 304 parameter is named `beta` inside the default brm model.

305 These estimates should be approximately 100 ms for Clear and 50 ms for Low-Blur.

306 The corresponding contrast can be obtained using:

```

307 # Default model: Blur contrast in beta (the exponential component).
308 tau_difference_default <- estimate_contrasts(
309   fit_brms,
310   contrast = "Blur",
311   predict = "beta",
312   centrality = "mean"
313 )

```

#### 314 **Table 5**

315 *Blur contrast in the exponential component (beta) from the default brms model.*

|     | Contrast         | Mean  | 95% CrI        | pd    |
|-----|------------------|-------|----------------|-------|
| 316 | Low-Blur - Clear | -51.6 | [-86.1, -19.7] | 0.999 |

318 As we can see in [Table 5](#), the difference between the two conditions is approximately 50 ms.

#### 319 **Classical Ex-Gaussian Parameterization**

320 As we have shown above, the default parameterization gives you the expected RT of the entire  
 321 distribution. We can instead fit the model using the `{brms.exgaussian}` package, which implements the  
 322 classical ex-Gaussian parameterization. In this model,  $\mu$  represents the location of the Gaussian  
 323 component and  $\tau$  represents the mean of the exponential component.

324 We first define the custom family and Stan functions from the `{brms.exgaussian}` package:

```

325 # Custom family that reparameterizes brms so mu is the Gaussian location.
326 exg_stanvars <- exgaussian2_stancode() # Stan code injected into the model
327 exg_family <- exgaussian2()           # the custom family definition

```

### 328 *Prior Specification*

329 The priors for the classical model are specified as follows:

```

330 priors_classical <- c(
331   # mu intercept: centered at 500 ms, now the Gaussian location of the
332   # reference condition (not E(RT)).
333   prior(
334     normal(500, 100),
335     class = "Intercept"
336   ),
337   # Effect of Blur on mu (Gaussian location): centered at 0.
338   prior(
339     normal(0, 50),
340     class = "b"
341   ),
342   # tau (exponential mean) intercept, log scale: exp(log(100)) = 100 ms.
343   prior(
344     normal(log(100), 0.5),
345     class = "Intercept",
346     dpar = "tau"
347   ),
348   # Effect of Blur on tau (log scale): centered at 0.
349   prior(
350     normal(0, 0.5),
351     class = "b",
352     dpar = "tau"
353   ),
354   # sigma (Gaussian SD) intercept, log scale: exp(log(60)) = 60 ms.
355   prior(
356     normal(log(60), 0.5),
357     class = "Intercept",
358     dpar = "sigma"
359   )
360 )

```

361 These priors follow the same reasoning as those for the default model (described above), with two  
 362 changes. First, the intercept is now centered at 500 ms rather than 600 ms, because in the classical  
 363 parameterization mu denotes the Gaussian-location parameter (500 ms for the Clear reference condition)  
 364 rather than the expected RT:

$$\mu_{\text{Intercept}} \sim \mathcal{N}(500, 100).$$

365 Second, because this model estimates  $\sigma$  explicitly, it receives its own weakly informative prior on the  
 366 log scale, centered at the simulated Gaussian standard deviation of 60 ms:

$$\log(\sigma) \sim \mathcal{N}(\log(60), 0.5).$$

367 The coefficient for Blur on  $\mu$  and the exponential-component priors are unchanged from the default  
 368 model; the latter now simply attach to  $\tau$  rather than  $\beta$ , reflecting the difference in labels rather than  
 369 any difference in the underlying quantity.

370 The model is then fit using:

```
371 fit_brms_new <- brm(
372   bf(
373     rt ~ Blur,      # mu submodel: here mu IS the Gaussian location
374     tau ~ Blur,     # exponential mean (the tail), allowed to vary
375     sigma ~ 1       # constant Gaussian SD
376   ),
377   data = d,
378   prior = priors_classical,
379   family = exg_family,      # custom classical ex-Gaussian family
380   stanvars = exg_stanvars,  # supplies its Stan implementation
381   chains = 2,
382   iter = 2000,
383   warmup = 500,
384   backend = "cmdstanr",
385   cores = 4,
386   seed = 1234
387 )
```

### 388 *Estimating the Gaussian-Location Parameter*

389 For the classical model, estimates from the primary regression formula correspond to the  
 390 Gaussian-location parameter  $\mu$ :

```
391 # Classical model: per-condition mu, now the Gaussian-location parameter.
392 mu_estimates <- estimate_means(
393   fit_brms_new,
394   by = "Blur",
395   predict = "mu",
396   centrality = "mean"
397 )
```

**Table 6**

*Estimated Gaussian-location parameter ( $\mu$ ) by condition from the classical model.*

| Blur     | Mu    | 95% CrI        | pd |
|----------|-------|----------------|----|
| Clear    | 476.7 | [452.1, 503.3] | 1  |
| Low-Blur | 544.7 | [524.5, 564.7] | 1  |

The estimates are approximately 500 ms for Clear and 550 ms for Low-Blur.

The contrast between the two can be obtained using:

```
# Classical model: Blur contrast in the Gaussian-location parameter mu.
mu_difference <- estimate_contrasts(
  fit_brms_new,
  contrast = "Blur",
  predict = "mu",
  centrality = "mean"
)
```

**Table 7**

*Blur contrast in the Gaussian-location parameter ( $\mu$ ) from the classical model.*

| Contrast         | Mean | 95% CrI      | pd |
|------------------|------|--------------|----|
| Low-Blur - Clear | 68   | [39.2, 95.8] | 1  |

The posterior mean will not necessarily equal exactly 50 ms because of sampling variability.

However, the 95% credible interval should contain the population value of 50 ms when the model successfully recovers the simulated effect.

### ***Estimating the Exponential Component***

Condition-specific estimates of  $\tau$  can be obtained by requesting predictions for the tau distributional parameter:

```
# Classical model: per-condition tau (the exponential component).
tau_estimates <- estimate_means(
  fit_brms_new,
  by = "Blur",
  predict = "tau",
  centrality = "mean"
)
```

**Table 8**

*Estimated exponential component ( $\tau$ ) by condition from the classical model.*

| Blur     | Mean  | 95% CrI       | pd |
|----------|-------|---------------|----|
| Clear    | 104.0 | [76.2, 136.2] | 1  |
| Low-Blur | 58.2  | [38.5, 81.6]  | 1  |

The estimates should be close to 100 ms for Clear and 50 ms for Low-Blur.

The corresponding contrast can be obtained using:

```
# Classical model: Blur contrast in tau (the exponential component).
tau_difference <- estimate_contrasts(
  fit_brms_new,
  contrast = "Blur",
  predict = "tau",
  centrality = "mean"
)
```

**Table 9**

*Blur contrast in the exponential component ( $\tau$ ) from the classical model.*

| Contrast         | Mean  | 95% CrI      | pd    |
|------------------|-------|--------------|-------|
| Low-Blur - Clear | -45.8 | [-77.3, -14] | 0.997 |

The results seen in [Table 9](#) are very similar to the default parameterization.

This example illustrates the central difference between the two parameterizations. The default `{brms}` ex-Gaussian model directly estimates the expected response and labels that parameter  $\mu$ . The exponential component is labeled  $\beta$ , although it corresponds to the classical  $\tau$  parameter. In contrast, the `{brms.exgaussian}` model uses  $\mu$  for the Gaussian-location parameter and  $\tau$  for the exponential mean, consistent with the classical parameterization commonly used in reaction-time research.

**Real Example****Conclusion**

With this commentary, I hope to encourage researchers to use the appropriate ex-Gaussian parameterization when modeling reaction-time data with the `{brms}` package. The ex-Gaussian distribution provides a powerful framework for moving beyond analyses of the mean by separating differences in the central portion of the reaction-time distribution from differences in its right tail.

However, this flexibility also requires careful attention to model parameterization. In particular, researchers should not assume that parameters with familiar labels necessarily have the same interpretation across software implementations. Researchers should therefore examine the statistical



definition and underlying implementation of a model before fitting it and interpreting its parameters. This is especially important when using flexible modeling software that may adopt parameterizations that differ from those commonly used in a particular research literature. I hope this commentary serves both as a practical guide to fitting the classical ex-Gaussian model in {brms} and as a broader reminder that convenient software defaults should not replace careful consideration of what a model actually estimates.

### References

- Angele, B., Baciero, A., Gómez, P., & Perea, M. (2023). Does online masked priming pass the test? The effects of prime exposure duration on masked identity priming. *Behavior Research Methods*, 55(1), 151–167. <https://doi.org/10.3758/s13428-021-01742-y>
- Angele, B., Gutiérrez-Cordero, I., Perea, M., & Marcet, A. (2024). Reading(,) with and without commas. *Quarterly Journal of Experimental Psychology*, 77(6), 1190–1200. <https://doi.org/10.1177/17470218231200338>
- Baayen, R. H., & Milin, P. (2010). Analyzing reaction times. *International Journal of Psychological Research*, 3(2), 12–28. <https://doi.org/10.21500/20112084.807>
- Balota, D. A., & Yap, M. J. (2011). Moving beyond the mean in studies of mental chronometry the power of response time distributional analyses. *Current Directions in Psychological Science*, 20(3), 160–166. <http://cdp.sagepub.com/content/20/3/160.short>
- Balota, D. A., Yap, M. J., Cortese, M. J., & Watson, J. M. (2008). Beyond mean response latency: Response time distributional analyses of semantic priming and word frequency effects. *Journal of Memory and Language*, 59(4), 495–523. <https://doi.org/10.1016/j.jml.2007.10.004>
- Bezanson, J., Edelman, A., Karpinski, S., & Shah, V. B. (2017). Julia: A fresh approach to numerical computing. *SIAM Review*, 59(1), 65–98. <https://doi.org/10.1137/141000671>
- Carpenter, B., Gelman, A., Hoffman, M. D., Lee, D., Goodrich, B., Betancourt, M., Brubaker, M. A., Guo, J., Li, P., & Riddell, A. (2017). Stan: A probabilistic programming language. *Journal of Statistical Software*, 76(1), 1–32. <https://doi.org/10.18637/jss.v076.i01>
- Dewitt, S. H., Lagnado, D., & Fenton, N. (2018). *Updating prior beliefs based on ambiguous evidence* (., M. R. J. Z. Chuck Kalish & T. Rogers, Eds.; p. 306311).
- Geller, J., Gomez, P., Buchanan, E. M., & Makowski, D. (2025). A distributional response time analysis of the perceptual disfluency effect. *Journal of Cognition*, 8(1), 50. <https://doi.org/10.5334/joc.469>

- Kane, M. J., & Engle, R. W. (2003). Working-memory capacity and the control of attention: The contributions of goal neglect, response competition, and task set to stroop interference. *Journal of Experimental Psychology: General*, 132(1), 47–70. <https://doi.org/10.1037/0096-3445.132.1.47>
- Kinoshita, S., & Liong, G. (2023). Mirror letter priming is rightward-biased but not inhibitory: Little evidence for a mirror suppression mechanism in the recognition of mirror letters. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 49(10), 1523–1538. <https://doi.org/10.1037/xlm0001239>
- Matzke, D., & Wagenmakers, E.-J. (2009). Psychological interpretation of the ex-gaussian and shifted wald parameters: A diffusion model analysis. *Psychonomic Bulletin & Review*, 16(5), 798–817. <http://link.springer.com/article/10.3758/PBR.16.5.798>
- Pfadt, J. M., Bartoš, F., Godmann, H. R., Waaijers, M., Groot, L., Heo, I., et al. (2025). A methodological metamorphosis: The rapid rise of bayesian inference and open science practices in psychology. *PsyArXiv*. [https://doi.org/10.31234/osf.io/ck3js\\_v1](https://doi.org/10.31234/osf.io/ck3js_v1)
- R Core Team. (2024). *R: A language and environment for statistical computing*. R Foundation for Statistical Computing. <https://www.R-project.org/>
- Ratcliff, R., & Murdock, B. B. (1976). Retrieval processes in recognition memory. *Psychological Review*, 83(3), 190–214. <https://doi.org/10.1037/0033-295X.83.3.190>
- Serrano-Carot, M., Gomez, P., & Angele, B. (2026). Articles as flankers: The effect of grammatical gender depends on the task. *Language, Cognition and Neuroscience*, 41(5), 676–698. <https://doi.org/10.1080/23273798.2026.2657830>
- Van Rossum, G., & Drake, F. L. (2009). *Python 3 reference manual*. CreateSpace.
- Vazire, S. (2018). Implications of the credibility revolution for productivity, creativity, and progress. *Perspectives on Psychological Science*, 13(4), 411–417. <https://doi.org/10.1177/1745691617751884>
- Whelan, R. (2008). Effective analysis of reaction time data. *The Psychological Record*, 58(3), 475–482. <https://doi.org/10.1007/BF03395630>
- Wilcox, R. R., & Keselman, H. J. (2003). Modern robust data analysis methods: Measures of central tendency. *Psychological Methods*, 8(3), 254–274. <https://doi.org/10.1037/1082-989X.8.3.254>